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Basics

This web site is somewhat daunting for a *new to mobile* operator, especially if one tries to take it all in at once. What's more, the articles are arranged alphabetically, not chronologically. As a result, one needs to know where to start, and this article is an attempt to set the logical sequence. The following verbiage is in a first-person writing style, and the reason will become apparent.

Let me say from the onset, that I am biased for and against some products. This is especially true when it comes to antennas, and the way they're mounted. Unfortunately, too many newcomers just can't get over the *stigma* of drilling a hole(s) to properly mount an antenna(s). Lame excuses aside, if you want the best performance (efficiency), drilling a hole is mandatory!

Just as important is the selected transceiver, how and where they're mounted, and how they're wired! Do it all correctly and enjoy the ride! Or, do it all haphazardly, and be plagued with issues, most of which you'll never realize you had!

All of this points out one indelible fact. Getting in a hurry to go mobile, is a prescription for spending more money than you *ever* intended, and having to do the job over! Maybe more than once!

If you're installation includes HF coverage, and you drive a hybrid or full-on electric vehicle, change your plans! If you want to know why, read the highlighted article.

Aluminum-bodied vehicles require special considerations with respect to wiring, antenna mounting, and bonding, to prevent *galvanic corrosion*. Ford's F150 series is a typical example, and Ford has issued a bulletin covering their offerings. While other aluminum-bodied vehicles may have similar guidelines, it is imperative to contact dealer and/or factory support personnel for specific details on other makes and models.

Lastly, taking any one comment, statement, or condition, out of one context, and applying it to another, will *always* give you *the wrong answer*! A good example of this, is the term *ground*! In other words, one form of ground, doesn't mean it is another form of ground. And, as pointed out in the highlighted article, the hardest of all the various grounds to understand, is the idea of what *is*, or what *isn't*, a ground plane. One thing for sure, calling it a counterpoise gets you off on the wrong foot!

Step One

If you haven't purchased a mobile transceiver, *don't!* At least until you have investigated all of the aspects of your intended selection. And just because your best pal bought a specific model, doesn't mean you should, as everyone's taste and take is different.

The article on Miniature Radios gives you the basic knowledge you need for HF transceiver selection, and the VHF Options article covers the VHF ones. And, if you're into VHF, don't buy a handheld as a first transceiver, and expect to use it as a mobile! If you do so anyway, it won't take you long to figure out why it isn't (wasn't) a good idea. Fact is, unless you already have an HF capable transceiver, don't buy an FM only one! The reason will become evident sooner than you might think!

While I prefer Icom products, those from Alinco, Kenwood, Yaesu, and others are just as good, if they have the features you're interested in. The number of built-in memory channels is a typical, must-have feature, but shouldn't be the main selection criteria. Remember too, in a mobile scenario, *selectivity* is much more important than *sensitivity*. This fact rules out every handheld on the market!

Almost all modern transceivers come with DSP (digital signal processing). Those with IF-based DSP (Icom IC-7000, IC-7100, IC-7300, and Yaesu's FT991) are preferred, over AF-based (Kenwood TS-480 or Yaesu FT-857) ones for several reasons. Let me add, that after-market, AF-based DSP units are very good, and sometimes better than built-in ones. But don't buy one just because, at least until have tried the built in one.

Typically, IF-based DSPs have adjustable bandwidths, and do not require additional IF filters (selectivity) to be installed. Transceivers without an IF DSP, will require purchasing narrower SSB filters in most cases. You might get by without doing so, but you'll soon realize why narrow bandpass filters are a basic requirement of mobile operation. The cost of these filters need to be considered in your purchase criteria.

One very valuable feature is *auto off*. And make sure you set it up during the installation process, lest you end up with a dead battery! And forget about ignition-switch-controlled relays, as they cause more issues than they solve. Incidentally, external automatic on/off devices have their place, but they're a poor substitute for built in ones.

If you think your new HF mobile transceiver *must have* a built-in antenna tuner, or your purchase plan includes an external one, make sure you read Step Two!

Older model, remotely mountable transceivers like the Icom IC-706MkIIg and IC-7000, or Yaesu FT100D, are no longer made. Thus finding remote kits for them is difficult at best. In some cases, manuals, power cables, accessories, bandpass filters, and finals, are hard to find or nonobtainium. Caveat Emptor is the phrase of want! Finals for these outmoded transceivers can also be an issue, which limits the appeal of used, on-line bargains!

So to aid the first time buyer, here are the salient points:

Don't put any faith in on-line reviews!
Don't consider any transceiver older than 10 years, unless you know repair parts are available, and you can do the job yourself!
Don't buy any transceiver you can't play with first! This eliminates auction sites as a source.
Don't buy any transceiver that has been modified in any way, shape or form!
Be very leery of bundled deals, particularly transceiver/antenna combinations.
*Built-in auto-couplers are **not** a basic requirements for any mobile transceiver!*
Buy a transceiver which is remotable. The reasons should be obvious!

Step Two

Without any doubt, *the single most important mobile purchase* you'll ever make is the ***antenna***! If you haven't already purchased one, *don't!* At least until you've read the highlighted article. If you don't wish to wade through the technicalities, here's the bottom line.

Remotely-tuned, HF mobile antennas (often called screwdrivers) are the prudent choice. They can be re-tuned while underway, and in most cases the function can be automated. This makes them a very attractive choice over monobanders like hamsticks, and other manually-tuned antennas. Some of the better choices, like the Scorpion SA-680, are very efficient, big, heavy, rugged, and long lasting. Thus they require due diligence in mounting, and often require home brewing mounting brackets. They often come with cap hats which further increases efficiencies. They are the breakfast of champions! Some folks say they're ugly, but form always follows function!

If your sights are a bit lower, Tarheel and Ameritron make some lessor performing models. In some cases, they too require fashioning special brackets, and drilling holes. And know this; Using a trailer hitch mount to avoid drilling a hole will reduce your ERP (Effective Radiated Power) by at least 50%!

Whatever you select, shy away from models which use a Polyfuse® or interrupted thread drive screw to protect the motor, as they're incompatible with all automatic antenna controllers.

Auto-coupler/whip combinations are a trade-off of efficiency versus convenience. They're essentially a based-loaded antenna, so their efficiency is about half that of a center-loaded

antenna of the same electrical length. In most cases, 80 and 160 meters are excluded. It should be noted that built-in, and some external auto-couplers do not have the range to match a 102 inch whip on any band other than 10 and 12 meters. If in doubt, read the high-lighted article.

If your sights are *really low*, then think about a short stubby, screwdriver antenna. They're often chosen because they're easy to mount (clip, lip, and magnets, etc), typically don't require drilling holes, and easily garner spousal approval. These exact attributes also make them *very lossy* indeed. You can work DX while using one, but only when band conditions are favorable. And, due mainly to the mounting styles often used, common mode becomes a *really* big problem. If you choose a short stubby antenna, understand they have *severe* limitations!

I'm not against monoband antennas like hamsticks and Hustlers. They work about as good or a bit better than a short stubby screwdriver, but their bandwidth is rather narrow especially on the lower bands. This can be a very limiting factor if you change bands as often as most mobile operators do.

Antennas advertised as having automatic band switching and/or requiring an external tuner to make them work, are nothing short of a scam! Remember, if it sounds to good to be true, it probably is!

If there is but one common thread among any of the above antenna choices, it's the need for an adequate ground plane. I know (not suspect) that far too many newbie amateurs don't have a clue what the difference is between a DC ground, an RF ground, and a ground plane! If you read the highlighted article, you'll get a feel what is what. You'll also discover that a ground strap to the nearest hard point doesn't replace, or act as, a ground plane. Here is the one thing every mobile operator needs to understand, and appreciate: *It is the (size of the) metal mass* ***directly under the antenna****, not what's along side,* ***that counts!***

What the last paragraph alludes to, is correct mounting. This almost always requires drilling holes in sheet metal. There are ways around the issue, but patience and perseverance are the key words. Luggage racks, rear hatches, and tire carriers are poor mounting locations. So are frame extensions if the ground plane isn't properly established. And contrary to popular belief, drilling a hole in sheet metal isn't the end of the world, even if you lease your vehicle!

By the way, all decent HF mobile antennas require a matching network. Using a built in tuner for this need, is wrought with operational problems especially when used with a screwdriver antenna. Further, if your antenna doesn't require matching to achieve an SWR under 1.6:1, you need a better antenna, a better mounting scheme, or both!

All too often, justification of one's mobile antenna installation is justified by quoting how many DX contacts have made with the antenna in question, a rather inane and trite practice! Here is why. The average efficiency of an 80 meter mobile antenna installation will be about 1%. That relates to 100 watts in, and one watt out! Yet, it is easy to make mobile to mobile contacts, even DX ones, on 80 meters.

Whatever choice you make, here are a few salient points to remember:

Don't even ***think*** *about buying a Yaesu ATAS! It is the lossiest, most trouble-prone, remotely-tuned antenna money can buy.*
Don't buy an antenna with a coil larger than 3.5 inches.
Don't buy an antenna with a coil smaller than 1.75 inches.
Don't buy an antenna with large metal end caps.
Don't buy an antenna which requires an external tuner, or advertised as automatic tuning!
Any antenna which ***doesn't*** *require matching, will be much lossier than one that does.*
Any antennas mounted on clip, lip, and mag mounts, will be lossier than permanently (through hole) mounted ones!
All antennas, especially ones mounted on clip, lip, and mag mounts, will require an extra-high impedance common mode coax choke, bar none!
All remotely-tuned (motorized) antennas will require an adequately-sized motor lead choke, bar none!
Overall length matters! Radiation resistance (efficiency) is directly related to the square of the electrical length. Thus, the longer the antenna, the more efficient it is!
The ability to work any given station, DX or otherwise, with any given antenna installation, is not indicative of any antenna-related parameter, including and especially efficiency!

Step Three

Installing amateur radio gear in and on a vehicle should never be taken lightly, but far to many do just that. Safety, must be ***paramount***! This means radio gear needs to be installed permanently, not stuck on! Magnets, wedged in blocks of wood, spring clamps, Velcro, or anything less than a bolt is not safe in the event of a vehicle crash! If you have to resort to these abbreviated means of attachment, you haven't planned your installation well enough! There is always a way to permanently attach radio gear without damage to any interior part. Flexible gooseneck mounts are one example of a no-hole mounting style.

Gear should be mounted to avoid as much distraction as possible. Remember, driver distraction is statistically 7 times more deadly than drunk driving!

Supplemental Restraint Systems (SRS or airbag for short) virtually cover the whole dashboard area, especially the top of it. Remember, airbags literally explode, and will toss asunder anything in their way, even *you* if you're not wearing a seat belt!

Cubby holes are very poor places to mount transceivers due to a lack of ventilation. They're passable for mounting control heads as long as transceiver controls and microphone cords don't interfere with vehicle controls. When mounting the main body of a remotable transceiver, always use the factory mounting bracket as this allows for better cooling. In other words, never mount a transceiver directly against carpeting! And forget about that extra heat-sink fan your buddy told you about.

Obviously, power and extension cables should always be routed to avoid personal contact by any passenger, including the driver. If they need to be longer than stock (≈10 feet), they should be rebuilt with properly sized wire to limit voltage drop (see step four below).

All of this seems a bit daunting, but trust me on this: There is always a way to safely mount radio gear, even though it may take hours to do so. Speaking of which, don't expect to do a complete install in one afternoon. The installation in my Honda Ridgeline took almost 30 hours, and I bet I'm a faster at it than most amateurs. It also helps to have a service manual for your specific model (in print, DVD-ROM, or On-Line). They're worth every penny you spend when you need to know what's behind a panel and/or how to remove the panel if required.

So the important points here are:

Don't mount anything atop the dashboard! Think airbag!
Don't mount gear in the foot wells, especially the driver's!
Don't get into a hurry! An average installation can take well over a day, and perhaps several days!
If you can, buy a Service Manual for your specific vehicle! Remember, it is often necessary to remove a seat or trim panel to route wiring.
Plan carefully, then follow your plan!
No magnets, bungee cords, wedged in block of wood, double-sided sticky tape, or Velcro!
Execute your plan diligently with ***safety*** *in mind.*
Be as neat and tidy as you can. Dangling wires are a distraction and a safety hazard!

Step Four

Proper wiring is just as important as any of the other steps, especially when dealing with aluminum-bodied vehicles. If there is but one rule to follow when wiring radio gear, it is this: ***Never, ever***, use existing vehicle wiring to power any radio gear! This includes, but is not limited to, so-called accessory sockets.

The common excuse for improper wiring is not finding an easy pathway through the firewall—*there is always a way!* If you purchased a repair manual as suggested above, you'd already know where to look! What's more, purchasing a separate battery to power the transceiver, however it is recharged, is just as inane as using existing wiring. Further, using that factory-supplied, 120 volt inverter as power or charging source, is asking for trouble. They're also a big RFI source!

Every transceiver manufacturer, and every vehicle manufacturer, has heretofore recommended direct to battery connections (both positive and negative leads), or to so-called jumps points when the battery is indirectly mounted under a fender well, or trunk area. However, that has *changed*! All late model vehicles, especially those with EIS (Engine Idle Shutoff),

have a battery monitoring system. Typically, these systems use a Hall device to measure battery current drain (Electronic Load Detector). However the wiring is done, it *must not* circumvent the ELD. A service manual is essential in these cases (in print, DVD-ROM, or On-Line). As an alternative, your dealer's service manager is a good resource. In any case, *both leads* should be fused as close to the power source (SLI battery) as possible.

Sometimes, it is necessary to extend factory-supplied power cables. It is important to keep the voltage drop through the wiring to less than .5 volt. If you don't understand why, then please read the wiring article for complete details.

Perhaps out of place, here is something to keep in mind. Almost without exception, there will be additional hardware required to complete any installation. Power Pole connectors, wiring lugs, sheet metal screws, crimp tool, and soldering iron are all good examples. It is impossible to set a specific cost for these items, but a good rule of thumb is to allow a plus 10% allowance for unforeseen circumstances. So, if you bought a $500 antenna, and a $1,300 transceiver, that's $180 for things you forgot you'd need when you planned your install!

Here is the bottom line for wiring:

Never, ever *use existing wiring to power amateur radio gear, especially accessory sockets!*
Proper fusing is essential, and both the positive and negative leads should be fused, no matter how they are connected!
Don't scrimp! Electrical fires are very costly indeed!
Keep voltage drop to a minimum! Less than .5 volts is the rule of thumb!
Trim or shorten excess cable lengths! This includes power wiring, and coax cables!
Be very neat in routing power wiring to avoid distraction, and interference with vehicle controls!

Last Words

One beginner aspect which should undergo a long and thorough thought process before indulgence, is running high power (>200 watts PEP). It is not uncommon to spend over $3,000 to buy the necessary hardware, which may include special brackets, and heavy-duty wiring. There are many reasons why this is so. Antennas must be *truly* capable of handling the higher power, and mounting always requires drilling holes in sheet metal! As the old cliché goes, it is always best to get your feet soaking wet before jumping into the deep water!

My personal motto is, *Set the example, don't be one!* Unfortunately, not everyone shares the sentiment. Whether you do or not, is immaterial. But ponder this: Why would anyone buy a $500 antenna, a $1,300 transceiver, and do a $5 installation? Beats me, but I see it all the time!

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